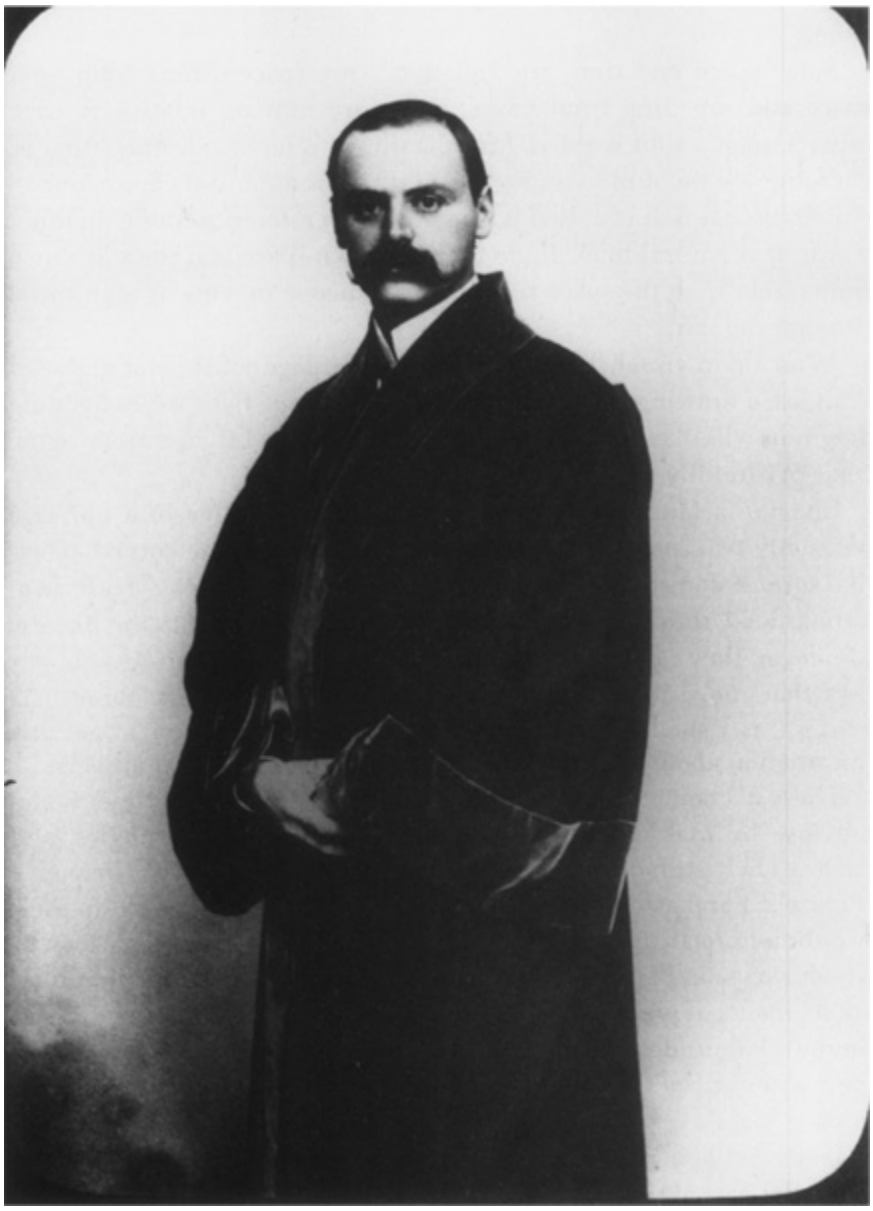


Karl Schwarzschild in his academic robe in Göttingen, Germany. [Courtesy AIP Emilio Segré Visual Archives.]

Since space and time are “relative” (my space differs from your space and my time from yours, if we are moving relative to each other[2]), such a split requires first choosing a reference frame—that is, choosing a state of motion. For a star, there is a natural choice, one in which the star is at rest; that is, the star’s own. reference frame. In other words, it is natural to examine the star’s own space and the star’s own time rather than the space and time of someone moving at high speed through the star.